

THE EVOLUTION OF PORTUGAL'S HIGH-TECH INDUSTRIAL POLICY FROM A NATIONAL INNOVATION SYSTEM PERSPECTIVE: A RESEARCH FRAMEWORK

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ABSTRACT

Grounded in the theoretical framework of the National Innovation System, this study develops an analytical model to examine the evolution of Portugal's high-tech industrial policy. The research divides the trajectory from the 1960s to the present into three stages: the foundation of infrastructure, the construction of innovation networks, and the recent focus on digitalization and green transformation. It further analyses the shifting dynamics from government-led initiatives to multi-actor collaboration. The findings suggest that Portugal has shaped an adaptive innovation system by embedding itself within the European Union framework and responding to global paradigm shifts. This study provides both a theoretical lens and empirical insights for understanding how medium-sized countries formulate innovation policies under conditions of globalization.

Key Words: National Innovation System theory; Portugal; high-technology industrial policy; EU science and technology policy

INTRODUCTION

Amid the global high-tech wave reshaping industry and competition, understanding how small- and medium-sized developed economies embed into global innovation chains through industrial policy has clear academic and practical value. Portugal – a mid-sized EU economy – illustrates a shift from peripheral status to emerging strengths in the digital economy and renewable energy, attracting substantial foreign investment and technology firms [1; 2]. This trajectory offers a salient case for innovation-driven development strategies. Existing scholarship provides multi-faceted views: sectoral studies on fintech, renewables, and digitalisation, and policy-institutional analyses on technology foresight [3], technology transfer [4], and sectoral planning and network governance [5]. Yet findings are fragmented across industries or intervals and lack a systematic account of Portugal's high-tech industrial policy since the 1990s, especially concerning the mechanisms of policy change.

To address this gap, the article advances an integrative framework grounded in National Innovation System (NIS) theory. NIS foregrounds coordination among firms, universities, research institutes, and the state within specific institutional settings [6; 7], offering analytic leverage to identify the systemic dynamics of policy transformation, including policy learning and path dependence. The study asks: (1) Since the 1990s, what stages has Portugal's high-tech industrial policy traversed, and what are the defining priorities of each? (2) Which factors drive these shifts, and how have NIS actors, together with international contexts – notably European integration – co-shaped the trajectory of policy evolution?

THEORETICAL FRAMEWORK AND LITERATURE REVIEW

Analytical Framework: The National Innovation System and Its Applicability. This study employs the National Innovation System (NIS) as the core framework. Originating in Freeman's analysis of Japan's catch-up, NIS conceives innovation as a systemic, policy-shaped process involving firms, universities, research institutes, government agencies, and intermediaries, rather than isolated firm activity. Lundvall's contributions further stress "interactive learning" and "knowledge flows" as central to innovation [7].

The framework synthesizes three analytical dimensions: (1) actor embeddedness in networks – innovation arises from collaborative, networked relations [8: 20–21]; (2) institutional embeddedness – policies, legal regimes, and cultural settings structure incentives; and (3) multi-level interactions – national systems interface with regional innovation systems and global networks in complex ways [9]. This lens suits Portugal, a mid-sized economy deeply integrated into the EU. Its policy choices are strongly conditioned by EU institutions and funding mechanisms [10]. Portugal's shift from technological catch-up to innovation-driven development reflects the ongoing coordination of internal NIS components [11] with external European resources. Applying NIS thus enables a systematic account of state roles,

coordination mechanisms, and multi-level governance in Portugal's policy evolution.

Literature Review: Progress and Limitations in Research on Portugal's High-Tech Industrial Policy. Portugal has become a notable European hub for financial technology, supported by an active entrepreneurial ecosystem. The literature splits into two strands. The first documents sectoral applications and current developments, detailing advances and attractiveness in fintech [12], renewable energy [13], and digitalisation/AI [1], alongside success in drawing foreign investment and technology firms [2]. These studies provide a solid empirical base. The second strand examines science and technology policy and institutional effects. Valadares Tavares analyzes the 2000 Technology Foresight (ET 2000) as a competitiveness-oriented agenda [3]. Maria das Dores and Teixeira compare technology-transfer policies [4]. Laranja shows how the Technological Plan (2005–2009) pivoted toward “network governance [5].” Additional work addresses R&D investment and industrial structure [14], entrepreneurship policies [15], and regional cases [16]. Significant gaps remain. Much research is fragmented – limited to single sectors, policies, or short periods – yielding no longitudinal, holistic account of Portugal's high-tech policy since the 1990s. Moreover, while studies like Laranja identify transitions, the systemic, multidimensional drivers of change are underexplored from an NIS perspective [5]. We still lack clarity on which interacting forces collectively produced paradigm shifts in innovation policy. Building on prior work, this article uses NIS to (a) delineate the phases of Portugal's high-tech industrial policy since the 1990s and (b) probe the mechanisms driving these changes, integrating theoretical application with empirical analysis.

THE EVOLUTION OF PORTUGAL'S HIGH-TECH INDUSTRIAL POLICY

Drawing on the NIS framework (actors–institutions–interactions), Portugal's high-tech industrial policy is a process of continuous adjustment of its national innovation system. The trajectory comprises three stages: an exploratory phase of system construction (1960s–1980s), a consolidation phase of systemic integration (1990s–early 2000s), and a transformation phase of systemic upgrading (2010s–present), each with distinct priorities and features.

Early Stage: Initial Construction of the NIS and Infrastructure. This period saw the emergence of modern S&T policy and the nascent NIS. Policy focused on core institutional frameworks and research infrastructure, with the state as principal driver and infrastructure builder. A key step was the creation of the National Board for Scientific and Technological Research

(JNICT) in 1967, enabling systematic S&T planning and management. The National Interim Development Plan (1965–1967) advanced industrial modernisation, productivity, and employment [17]. By the late 1980s–early 1990s, PEDIP (1988–1992) and the CIENCIA Programme (1990–1993) aimed to strengthen technological capabilities and reduce external dependence. From an NIS perspective, systemic coherence remained limited: firm–university linkages were weak, coordination shallow, and overall effectiveness constrained [11]. The NIS displayed early government-led, linear characteristics.

Middle Stage: Systemic Integration under EU Accession. EU accession was a watershed. EU structural funds and policy frameworks integrated domestic and external resources and built collaborative networks, shifting the NIS from stage-setting to orchestrating collaboration. Legally and strategically, the Law of Scientific and Technological Development (1988) and the Lisbon Strategy (2000) – including independent evaluations of research institutions – improved governance quality [10]. Operationally, the Interface Mission and PRODEX (ESA), largely EU-funded, and the launch of PROINOV promoted firm–university cooperation [15], signalling a turn from R&D subsidies toward networked innovation. R&D expenditure rose from 0.76% of GDP in 2005 to 1.58% in 2009 [14]. As Laranja (2012) notes, a shift to network governance became evident, with markedly enhanced coordination among actors.

Recent Stage: Systemic Upgrading towards Digital and Green Transformation. Since the 2010s, under Industry 4.0 and carbon-neutrality imperatives, policy has entered a phase of systemic upgrading driven by digitalisation and green transition. The focus moved from building collaboration to steering the system toward frontier technologies and alignment with the sustainable development agenda. The Indústria 4.0 Programme spans six domains – human resources, technological collaboration, start-ups, financing, internationalisation, and regulatory adaptation – pursuing a comprehensive innovation ecosystem. Complementary measures include ACP.2030 (advanced computing) and Portugal INCoDe.2030 (digital competences), strengthening digital infrastructures and human capital. On the green front, tax incentives, PPP mechanisms, and a long-term carbon neutrality commitment promote renewables and clean-tech innovation [13]. These shifts demonstrate adaptive responses to techno-economic change. System design now emphasises breaking sectoral silos, technological convergence, and deeper insertion into global innovation networks through international cooperation (e.g., EuroHPC JU), indicating a more mature, foresight-oriented, and open NIS.

DRIVING FORCES BEHIND THE EVOLUTION OF

PORTUGAL'S HIGH-TECH INDUSTRIAL POLICY

From an NIS perspective, policy evolution reflects joint internal–external influences. The drivers across stages centre on the institutional environment, actor interactions, and knowledge flows.

1960s–1980s: Domestic Modernisation and External Integration. Policy change was propelled by endogenous needs for economic upgrading and by pressures linked to integrating into the European Community. Portugal's reliance on low value-added industries and weak competitiveness made S&T investment a priority, prompting policies to strengthen national laboratories and industrial technological capacity. EC accession in 1986 supplied structural funds and exposed Portugal to higher standards and market competition, pushing alignment of the domestic S&T system with EU frameworks and accelerating modernisation of innovation infrastructure. Governance began to diversify beyond the central state, gradually involving local governments, universities, and firms – an early shift from a linear model toward exploratory networked governance. Overall, government acted as primary system builder through infrastructure investment and macro-planning. Yet limited firm–university interaction constrained synergies; knowledge flows were inefficient, and the system largely remained in catch-up and imitation mode [11].

1990s–Early 2000s: Systemic Deepening under the EU Framework. EU institutions became decisive, with structural funds and research programmes aligning national agendas to European objectives and steering activity toward international collaboration and university–industry partnerships. Responding to the global knowledge economy, policy tilted toward knowledge-intensive sectors to reduce dependence on traditional manufacturing. Policymakers also targeted regional imbalances by directing resources to inland regions (e.g., science parks) to narrow coastal–inland gaps. The state's role evolved from dominance to coordination and platform building, actively promoting knowledge flows among firms, universities, and research institutes. The NIS grew more open and networked, extending functions from basic support to value creation. Nonetheless, slow industrial restructuring and weak firm-level innovation capacity indicated that collaborative mechanisms still needed optimisation [10].

2010s–Present: Upgrading amid Global Technological Shifts. Global paradigm changes – digitalisation, intelligent technologies, and green transition – shaped a new phase oriented by technological foresight and internationalisation. To meet the Industry 4.0 race, Portugal adopted ENI 2030 to prioritise R&D in strategic domains. Competitiveness in frontier technologies became urgent, AI Portugal 2030 and ACP.2030 sought po-

sitions in AI and advanced/quantum computing, reinforcing technological sovereignty. The European Green Deal and domestic low-carbon goals moved green innovation to the fore, while participation in initiatives such as EuroHPC JU embedded the NIS in global knowledge networks and helped attract talent. The system has become more open, agile, and integrated, with the state acting as policy guide and ecosystem enabler through strategic direction, incentives, and international cooperation. Key challenges persist, such as strengthening indigenous innovation capacity in firms and improving allocation of innovation resources remains central to a mature NIS [13].

CONCLUSION

This article has sought to construct an analytical framework based on the theory of the National Innovation System (NIS) to interpret the evolutionary logic of Portugal's high-tech industrial policy. The framework conceptualises policy change as the dynamic evolution of an innovation system across three dimensions: the institutional environment, diverse actors, and interaction networks. Applied to the Portuguese case, the analysis traced the trajectory from infrastructural development to the construction of innovation networks and, more recently, to the prioritisation of frontier domains. It also demonstrated the adaptability of the NIS framework in explaining how medium-sized countries respond to the dual pressures of globalisation and technological transformation.

This research should be regarded as an exploratory effort in framework building. Its preliminary findings suggest that NIS theory provides a reasonable lens for understanding the dynamics of Portugal's STI policy evolution. The shifting role of the government, the growing synergy of innovation networks, and the strategic utilisation of knowledge resources at both the EU and global levels together depict the resilient development of an innovation system in a medium-sized country. Nevertheless, this remains only an initial framework, whose analytical depth and breadth warrant further expansion. The contribution of this study lies not merely in offering a narrative of the Portuguese case, but in proposing an analytical tool open to discussion, testing, and refinement. It is hoped that this framework will stimulate greater scholarly attention to the innovation strategies of medium-sized countries and foster both theoretical enrichment and practical exploration in the study of innovation policy dynamics under globalisation.

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